Surface Defects and Theories of class R:

Motivation: To study properties of coupled ~~2d-4d and~~ 2d-3d systems.

As we'll see in the 2d-4d case, the physics of the 2d system is modified ~~by~~ when we couple it to a 4d system. In some sense, we hope to find similar results in 2d-3d case.

Outline:

1). A (2,2) GLSM in 2d.

2). Surface Defects in 4d $\mathcal{N}=2$ theories, (class S, $T[C]$).

3). Domain walls in 4d $\mathcal{N}=2$ theories (class R, $T[M_3]$).

4). Surface Defects in Theories of class R.

1). A (2,2) GLSM in 2d:

(2,2) SUSY Multiplets:

1). chiral $\Phi \ni (\phi, F)$

2). Twisted chiral $\Sigma \ni (\sigma, D - iF_{01})$

A GLSM:

	$\bar{\Phi}_1$	$\bar{\Phi}_2$
$U(1)_g$	1	1
$SU(2)_f$	fund.	

with a 'twisted superpotential term' i.e. depends on twisted chiral Σ only $\int d^2\tilde{\theta}\, \tilde{w}(\Sigma)$

with $\tilde{w} = t\Sigma$.

for $t = -r + i\theta$; $r \in \mathbb{R}$, $\theta \in [0, 2\pi)$.

Also introduce twisted mass M for $SU(2)_f$: i.e. bkgd gauge super-field with scalar M but all the other fields set to zero.

M is diagonalizable. $\begin{pmatrix} m & 0 \\ 0 & -m \end{pmatrix} = M \in \mathfrak{sl}(2,\mathbb{C})$.

RG structure: r goes through dim. transmutation to give rise to Λ.

We ask: What is the LEET at $\mu \ll \Lambda$?

Witten:

Chirals are integrated out
And we get a theory of Σ_i.
with effective twisted superpotential

$$\tilde{W}_{eff}(\Sigma_i) = t\Sigma_i - \mathrm{Tr}\left((\Sigma_i + M)\log\left(\frac{\Sigma_i + M}{e}\right)\right)$$

The theory has discrete number of vacua, given by extremizing $\tilde{W}$.

$$\partial_\sigma \tilde{W} = 0$$

$$\Rightarrow \quad \boxed{\sigma^2 = e^t + m^2}$$

This equation is called twisted chiral ring. (TCR)
It represents the moduli space of vacua.
There are two vacua for each fixed value of t and m.

2). Surface Defects in 4d $\mathcal{N}=2$ theories:

Multiplets of $d=4$, $\mathcal{N}=2$:

Vector $\Psi \ni (\psi, F_{\mu\nu}, F_\psi)$

Chiral $\Theta \ni (\theta, \hat{\theta}, F_\theta, F_{\hat{\theta}})$

Purpose:

To show that TCR is modified.

Surface Defects are localized on a two-dim subspace $\mathbb{R}_{0,3} \subset \mathbb{R}_{0,1,2,3}$.

- Can preserve at most 4 supercharges. They make a (2,2) algebra in 2d sense.
- One can ask how do bulk superfields decompose into multiplets of (2,2)?

Answer: $\Phi \Big|_{\text{defect}} = \Gamma \ni (\gamma, \cancel{\nabla_{\parallel}}, F$

Modifies $S = \int d^4x \, \mathcal{L}_{4d}$ to $S = \int d^4x \, \mathcal{L}_{4d} + \delta^2(x_\perp)\mathcal{L}_{2d}$.

<u>An example:</u>

Take pure $SU(2)$ YM.

<u>Recipe for choosing $\mathcal{L}_{2d}$:</u> Take $\mathcal{L}_{2d}$ s.t.

FLAVOR OF DEFECT $SU(2)_f$

= GAUGE OF BULK $SU(2)$.

<u>Coupling procedure:</u> Gauge $SU(2)_f$!

ask what is LEET at $\mu \ll \Lambda_{4d}$?

First integrate out 2d chirals and modes of Σ_1:

$$\tilde{W}_{eff}(\Sigma_1, \Gamma) = t\,\Sigma_1 - \mathrm{Tr}\left((\Sigma_1+\Gamma)\log\left(\frac{\Sigma_1+\Gamma}{e}\right)\right)$$

Now integration over 4d fields yields

$$\tilde{W}_{eff}(\Sigma_1, \Gamma) = t\,\Sigma_1 - \left\langle \mathrm{Tr}\left((\Sigma_1+\Gamma)\log\left(\frac{\Sigma_1+\Gamma}{e}\right)\right)\right\rangle.$$

We don't know how to compute this exp. value but it is related to the <u>resolvent</u> of the 4d theory as follows.

$$\partial_\sigma^2 \tilde{W}(\sigma, \Gamma) = -\mathrm{Tr}\left\langle \frac{1}{\sigma+\Gamma}\right\rangle$$

$$= \frac{-2\sigma}{\sqrt{(\sigma^2-u)^2 - 4\Lambda^4}}$$

computed in (hep-th/0211170)

where $u = \mathrm{Tr}\langle\gamma^2\rangle$. $\left[\text{Semiclassically, } \gamma = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}.\right]$

Now integrate it once to get

$$\partial_\sigma \tilde{W} = t - \log\left(\frac{\sigma^2 - u + \sqrt{(\sigma^2-u)^2 - 4\Lambda^4}}{2}\right)$$

Extremize to get TCR:

$$\boxed{\sigma^2 = e^t + u + \Lambda^4 e^{-t}}$$

Rks:

1). TCR is modified by 4d instanton corrections.

2. $\boxed{\text{TCR} = \text{SW}}$

Claim: For any theory of class S, $\exists$ a S.D. whose TCR satisfies

$$\boxed{\text{SW} = \text{TCR}!}$$

Example 2:

Consider a 4d theory of a free massive hyp $\ominus$ of mass parameter m.

$$\Sigma_{SW} = \{ (x,z) \in \mathbb{C}^2 \mid x^2 = z^2 - m \}.$$

$m \in \mathbb{C}$

and mass of the hypermultiplet is $|m|$.

It has a $U(1)$ global symmetry. Call it $U(1)_{bulk}$.

Roughly, the following 2d theory has the ~~correct~~ right TCP.

	Φ_1	Φ_2
$U(1)_g$	1	-2
$U(1)_f$	1	0

where $U(1)_f$ is identified with the ~~restriction of~~ bulk flavor $U(1)$.

Call the S.D. Lagrangian $\tilde{\mathcal{L}}_{2d} \equiv \tilde{\mathcal{L}}_{2d}(\sigma, m)$ as it will be useful later.

3). Domain walls in 4d $\mathcal{N}=2$ theories:

Recipe: 4d $\mathcal{N}=2$ theories have following parameters:

$$f \begin{cases} \bullet\ \tau_i : \text{Complexified gauge coupling.} \\ \bullet\ m_j : \text{bare hyp masses.} \\ \bullet\ \Lambda_k : \text{QCD-like scales.} \end{cases}$$

Goal: To construct $(2+1)$d ~~defects~~ defects that preserve $\mathcal{N}=2$ SUSY in 3d sense.

~~Vaker~~ To build DW on $\mathbb{R}_{0,1,3} \subset \mathbb{R}_{0,1,2,3}$

take $f \in \{\tau_i, m_j, \Lambda_k\}$ $\quad \Psi(x^0, x^1, x^2, x^3)$

and vary with x^2 : $f = f(x^2)$.

This breaks 4d Poincare invariance to 3d Poincare invariance, and we can preserve at most 4 supercharges i.e. $\mathcal{N} = 2$ SUSY in 3d sense.

The Q – Q superalg comm. rel. in this 3d $\mathcal{N}=2$ subalgebra are as follows:

$$\{Q, Q\} \sim \mathrm{Re}\,(Z_{4d}).$$

Therefore we can diagonalize this subalgebra to find BPS condition: $M = \mathrm{Re}\,(Z_{4d})$.

but recall that central charges of 4d BPS states depend on the parameters $\{\tau_i, m_j, \Lambda_k\}$.

To make $M = \mathrm{Re}(Z_{4d})$ independent of x_2, we vary $f \in \{\quad\}$ s.t.

$$\boxed{\partial_2 \,\mathrm{Re}\,(Z_{4d}) = 0.}$$

~~4d BPS states will be trapped as 3d BPS states~~

We also impose that $|Z_{4d}(x^2)| \to \infty$ as $|x^2| \to \infty$.
So that BPS particles are very massive far out.

4d BPS states are <u>trapped</u> as 3d BPS states at points where $\mathrm{Im}(Z_{4d}) = 0$.

Cecotti-Cordova-Vafa (1110.2115):

~~3d system produced this way~~

~~To write a Lagrangian description of this system,~~

~~To describe the system in action formalism,~~

~~choose a Lagrangian such that it produces the~~

3d $\mathcal{N}=2$ Multiplets:

Example:

Consider a free massive hypermultiplet on $R^{1,2} \times S^1_{x^2}$, where the circle has radius R. There is only one parameter in this theory, the mass of the hypermultiplet. Let m vary with x^2 as

$m = m_0 + i\frac{1}{R}\tan^2\theta/2$. As $\theta \to \pi, -\pi$, the hypermultiplet becomes very massive, and at *theta* $= 0$, it reduces to a 3d BPS state as mentioned above. At small energies, this BPS state will look like its localized at $x_2 = 0$.

Cecotti-Cordova-Vafa propose that in such a situation, you may describe the system in terms of a 3d N=2 chiral multiplet of mass m_0, i.e. $L_{3d} = L(\chi)$. The chiral is charged under $U(1)_f \times U(1)_r$ with charges $(1,0)$. This is an example of the Theories of class R. In this case, $M = \mathbb{C} \times S^1$, and $\tilde{M} = \Sigma \times S^1$.

Domain walls in Theories of class R

Goal: We want to write down a surface defect inside the domain wall, such that it's vacuum equation coincides with $\tilde{M}$. That is, a Lagrangian of the form $L = L_{3d} + \delta(x_1)L_{2d}$.

The first question is: what is the maximum number of supercharges that this surface defect can preserve? Answer: 2. They fall into a (1,1) superymmetry algebra in 2d.

The second question is: how to write down a lagrangian of this system to produce desired qualities? Answer: Start with inserting a surface defect in the 4d theory of a free massive hypermultiplet with varying mass $m = m_0 + \frac{i}{R}\tan^2\theta/2$. Think of this mass as a scalar in a background vector superfield. Now as it couples to the surface defect inserted at $\theta = \theta_0$, it appears as a twisted mass with value $m_0 + i\tan^2\theta_0/2$. As we go to energies below 1/R, the system has a lagrangian of the desired form. To compute the twisted chiral ring, extremize the effective twisted superpotential. It amounts to getting the following result.

$$x^2 = z^2 - (m_0 + i\tan^2\theta_0/2) \tag{1}$$

Future directions: This in some sense is the most trivial example one can work with. There are no gauge fields in the bulk, and as we have seen that introducing bulk gauge fields particularly complicates the situation. So in future, we hope to work on following things. — surface defects in domain walls obtained through Λ or τ flow. — look at other geometries